

THE PLATFORM

One control core. Power conversion at every scale.

OpenVVVF separates a reusable dual-MCU control and safety core from the power hardware. Its PWM, protection, and sensing interfaces can connect to smaller user-supplied stages, the implemented Size 2 inverter, the released Size 2 DC/DC module, or future higher-voltage and higher-current chassis. The result is an open foundation for motor drives and other power-conversion systems without locking each application to a new controller.

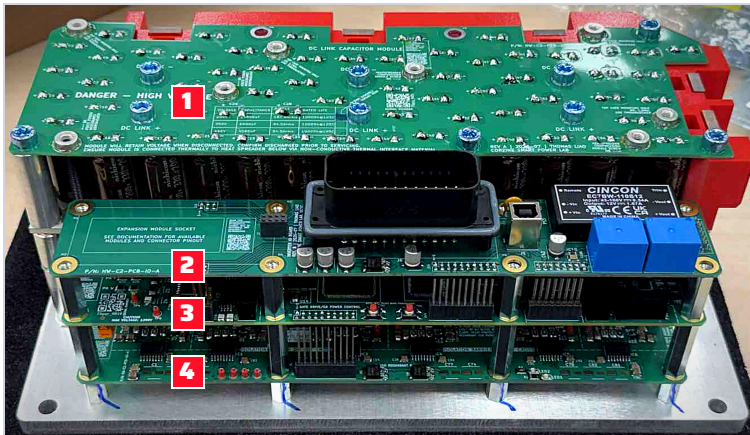


Fig. 1: The Size 2 stack, top to bottom: capacitor board, I/O board, control board, and gate-drive board.

1 DC-link capacitor board

Electrolytic cans are thermally bonded to the baseplate, with film and ceramic capacitors beneath. Multilayer ceramic capacitors (MLCCs) mount directly between the IGBT terminals. The short commutation loop limits switching overshoot and supports a future transition to SiC.

2 I/O board

USB-B, dual CAN FD, and an expansion socket for application I/O.

3 Control board, dual-MCU

FOC motor control runs on the main MCU. An independent safety coprocessor monitors every safety-critical signal and can bring the system to a safe state.

4 Gate-drive board

Isolated gate drive and sensing for the three IGBT half-bridges.

COOLING

Three IGBT half-bridge modules are conduction-cooled through an aluminum baseplate. Size 2 requires no liquid-cooling loop.

SOFTWARE AND WORKFLOW

The same control core is intended to run across power stages. It supports motor self-commissioning for PMSM and induction machines, the open-source RTE Studio model-based workflow, and the browser-based Telemetry Viewer. RTE Studio requires no MATLAB; a software-in-the-loop pipeline is in progress. The controls program has benchmarked ADRC and MPCC against the production FOC controller.

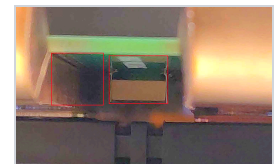


Fig. 2: Ceramic capacitors between the IGBT terminals. The commutation loop measures only a few millimeters.

MODULAR POWER BLOCKS

Scale down or up around one control core

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



Lower-power stages

Smaller or custom gate-drive stages with interface scaling

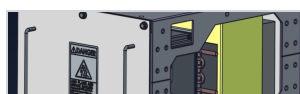
■ SCALABLE INTERFACE



Size 2 inverter

200–450 V class · 465 A continuous design rating

■ IMPLEMENTED, UNDER TEST



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

■ RELEASED DESIGN



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ IN DEVELOPMENT

EVIDENCE

Bench-tested, on the record

200-450 V Size 2 DC-bus class by capacitor selection	465 A Continuous design rating · 600 A peak (60 s)	10 Published test reports, raw telemetry linked	PMSM + induction Instrument-verified motor self-commissioning
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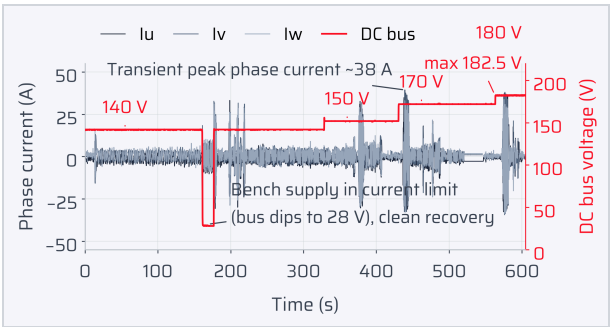


Fig. 3: 180 V power-stage bring-up. A Size 2 chassis (200 V-class DC link) drove a TECO MAX-IE3 10 HP induction motor as the bus stepped through 140, 150, 170, and 180 V. At approximately 160 s, the supply reached its current limit; after the limit was raised, the controller recovered without a reset or fault. Peak phase current was approximately 38 A; maximum bus voltage was 182.5 V.

TEST CAMPAIGN	SETUP	KEY RESULT
180 V power-stage bring-up	TECO 10 HP induction motor; bus stepped from 140 to 180 V	200 V-class operation confirmed; ≈38 A peak; clean fault recovery
One-hour no-load endurance	Size 2 + induction motor; 120 V DC; ≈60 Hz	60 minutes of continuous operation; zero faults; thermal baseline established
20-minute reversal stress at 180 V	±60 Hz every ≈45 s; passive cooling	26 reversals; zero faults; ≈41 A maximum
Self-commissioning accuracy	Inverter estimates versus LCR meter; PMSM + induction	Resistance error from -6.5 to +7.3 %; inductance error of +2.4 %

Six additional instrumented calibration reports support these results. All ten reports include raw telemetry and are revised publicly as validation continues.

SAFETY ENGINEERING

Six independent paths to the safe state

The control module is dual-MCU by design. The main MCU (**STM32H723ZG**) runs FOC motor control, plausibility checks, and fault handling. The safety coprocessor (**STM32G474RCTx**) independently monitors every safety-critical signal, cross-checks both CAN buses, and runs a challenge-response watchdog. Either processor can force the safe state.

Hardware PWM break (TIM1)	<100 ns	Shared driver RESET	<1 μs
1oo2 (one-out-of-two) gate-drive power kill	-10 μs	Independent coprocessor kill	<10 μs
Watchdog-forced main-MCU reset	-100 ms	Passive 3.3 V rail-loss pull-down	passive, no software

ASIL D design target for the highest safety goals, using ASIL B(D) decomposition: two independent ASIL B channels satisfy an ASIL D goal. ISO 26262 methodology is applied as a design input, not presented as a certification claim. The functional-safety roadmap includes fault-injection, thermal, and vibration campaigns.

17 hazards identified	>	14 safety goals	>	22 functional safety requirements	>	92 fault-injection tests planned	>	ISO 26262 + IEC 61800-5-2 safe-function mapping
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Safety engineered from the silicon to the sheet metal.

What sponsorship moves forward

OpenVVF is built in the Smart Power Lab at UC Santa Cruz. The lab's mission is to advance open power electronics by publishing hardware, firmware, and safety engineering that others can study, build, and extend.

The team publishes what it learns: schematics and PCB releases, control firmware, hazard and threat analyses, and the raw telemetry behind every test claim. Financial and in-kind sponsors provide the components, fabrication, instrumentation, and test access needed to turn open designs into validated hardware.

SUPPORTED BY



WHAT SPONSORSHIP FUNDS

■ IN DEVELOPMENT

Next control-module revision (Gen 7.1)

Hardware revision of the dual-MCU control module.

■ PLANNED

Loaded and dynamometer testing

Dynamometer testing of Gen 7.1 will take the platform from no-load validation to rated power.

■ IN PROGRESS

Zero DSR integration

A Size 2 chassis is being readied for our 2018 Zero DSR electric motorcycle.

■ IN PROGRESS

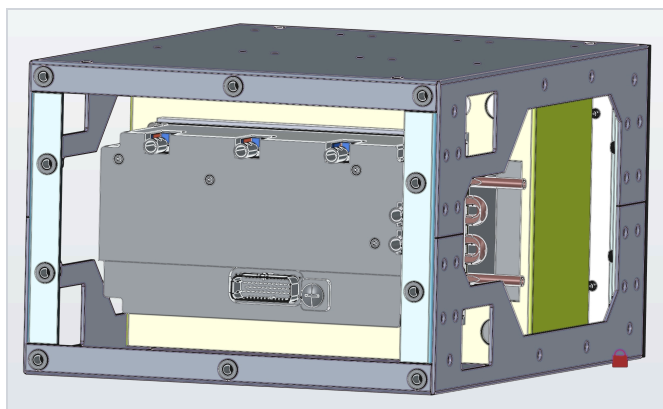
Modular DC/DC validation

The released 15 kW, 450 V-class module reuses the Size 2 boards and power stage. Sponsorship supports fabrication and load testing.

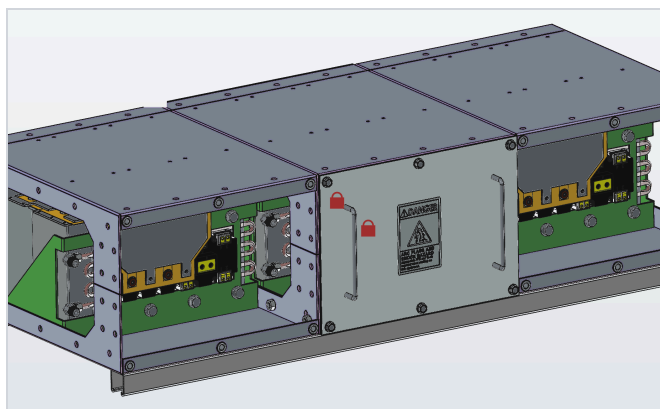
■ IN DESIGN

Size 3 high-power platform

The architecture extends to an up-to-1,200 V, 1,400 A chassis with a modular liquid-cooled enclosure. Arc-flash containment analysis is in progress.



Size 2 DC/DC module: Released design rendering. The 15 kW converter reuses the Size 2 PCBs and power stage; orderable BOMs are available.



Size 3 high-power enclosure: Design rendering. The up-to-1,200 V, 1,400 A concept uses modular liquid-cooled bays; arc-flash containment analysis is in progress.

Sponsor the next build and test cycle. Help move OpenVVF from bench evidence to loaded testing, vehicle integration, and reusable power-conversion modules.

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